Digestive Function, Fluid Loss, and Endocrine Signaling

NURSING · nursing-foundations · BoardPrep lesson v1

Official examination scope

The PRC references establish examination scope only. OpenStax and Open RN support general anatomy, physiology, and assessment education. The NHS stoma information is UK patient guidance with a review date that has passed and is limited to a caution about dehydration signs and contacting a qualified stoma professional. No local diagnostic threshold, fluid prescription, diet, medicine adjustment, or Philippine pathway is provided.

Learning objectives

- Trace digestion, absorption, and fluid balance through the gastrointestinal tract and relate
- them to a focused assessment.
- Recognize that persistent diarrhea or changed stoma output can accompany fluid/electrolyte
- risk and identify specific cues that should be assessed and reported.
- Explain endocrine signaling through hormones and target cells and distinguish physiology
- concepts from an individual diagnosis or treatment plan.

1. Relate digestive processes to absorption

The gastrointestinal tract performs linked processes: ingestion, propulsion, mechanical and chemical digestion, absorption, and elimination. The small intestine is the principal site of digestion and absorption of released nutrients; the large intestine has a later role in absorbing remaining water and electrolytes. These are related functions, not interchangeable labels: digestion breaks material down, while absorption moves selected products into blood or lymph. In a worked example, a question about where most nutrient absorption occurs points to the small intestine, whereas a question about residual water handling concerns the large intestine. An ostomy changes which segment empties through the stoma, so the expected output depends on anatomy and the individual’s baseline. This physiology helps organize assessment but cannot diagnose a cause, define an abnormal output threshold, or prescribe diet or fluid replacement.

Sources for this section

Section reference: Anatomy and Physiology 2e — OpenStax, Rice University — https://openstax.org/details/books/anatomy-and-physiology-2e

2. Assess gastrointestinal fluid-loss patterns

Bowel symptoms can affect fluid and electrolyte balance, so assessment should connect the person’s report with observed trends. Clarify stool or stoma pattern and change from baseline, intake, medication and illness history, dizziness or weakness, urine pattern, weight when available, vital signs, skin and mucous membranes, mental status, and relevant ordered laboratory results. Open RN identifies diarrhea as a potential cause of deficient fluid volume and supports interpreting intake/output, weight, vital-sign, skin, pulse, and mental-status findings together. For example, watery output with dizziness is not answered by looking at the stoma alone: assess the output trend and signs of volume change and communicate the findings to the responsible clinician or stoma service. A UK NHS page also connects high output, dry mouth, or fatigue with dehydration, but its stated review date has passed; currentness and local referral advice must be checked. Do not advise a specific oral fluid, salt, diet, medicine, or output threshold from this lesson.

Sources for this section

Section reference: 15.6 Applying the Nursing Process — Nursing Fundamentals 2e — Open RN; Wisconsin Technical Colleges System — https://wtcs.pressbooks.pub/nursingfundamentals/chapter/15-6-applying-the-nursing-process/
Section reference: Complications of an ileostomy — National Health Service, United Kingdom — https://www.nhs.uk/tests-and-treatments/ileostomy/complications/

3. Explain endocrine signaling and homeostasis

Endocrine tissues release hormones into extracellular fluid, from which signals travel through blood to cells with appropriate receptors. The response depends on the hormone, target cell, and physiologic context; some organs, including the pancreas, have both digestive and endocrine functions. Compared with neural signaling, endocrine effects are often longer-distance and usually slower, although timing varies and interactions can produce rapid responses. This is a functional distinction, not a rigid rule that every hormone acts slowly.
In a worked example, insulin and glucagon are pancreatic hormones that participate in blood-glucose regulation, while the pancreas also releases digestive secretions through ducts. A learner should keep those endocrine and exocrine functions distinct. This section teaches basic physiology, not interpretation of a specific laboratory value or a treatment regimen.

Sources for this section

Section reference: Anatomy and Physiology 2e — OpenStax, Rice University — https://openstax.org/details/books/anatomy-and-physiology-2e

4. Interpret endocrine cues without treating one value

A hormone result represents one piece of evidence; symptoms, timing, history, medications, and the relevant reference interval affect interpretation. OpenStax notes that suspected endocrine dysfunction may prompt additional testing and that treatment depends on diagnosis. For nursing assessment, compare the present pattern with baseline and the stated reason for testing, look for accompanying changes, verify when a sample or medicine was taken if relevant, and communicate uncertainty to the responsible clinician. A person with repeated out-of-range values and new symptoms needs a contextual review, not an inferred diagnosis from one number.
Do not change an insulin or hormone dose, establish a target, or provide a treatment recommendation from this lesson. Interpretation and treatment require qualified clinician review, current local guidance, and the actual patient record.

Sources for this section

Section reference: Anatomy and Physiology 2e — OpenStax, Rice University — https://openstax.org/details/books/anatomy-and-physiology-2e

Connect watery output with assessment

Question: A client with an ileostomy reports watery output and dizziness.
1. Clarify output pattern, timing, and change from the person’s usual output.
2. Assess intake/output and observable fluid-status cues such as dizziness, urine pattern, mucous membranes, pulse, and mental status.
3. Communicate the findings through the current local stoma-care or medical pathway; do not prescribe a fluid or diet plan from this lesson.
Answer: Assess the output trend and fluid-status context, then obtain qualified guidance when findings are concerning.
Check: No UK output threshold or fluid-replacement instructions are applied as Philippine protocol.

Separate digestion from endocrine function

Question: A learner is asked how the pancreas contributes to both digestion and glucose control.

1. Identify pancreatic digestive secretions that enter the GI tract through ducts.
2. Distinguish endocrine cells that release insulin and glucagon into circulation.
3. Relate each signal to its target and function without inferring an individual treatment.
Answer: The pancreas has both exocrine digestive and endocrine hormonal functions.
Check: The organ’s two functions do not make a digestive secretion a hormone or vice versa.

Review an isolated endocrine result

Question: A patient notices one abnormal hormone result and asks whether to change a medicine.
1. Do not infer the diagnosis or dose from the result alone.
2. Check the testing context and ask about symptoms or relevant changes.
3. Refer medication and interpretation questions to the authorized clinician and current plan.
Answer: A single result needs context and qualified interpretation; no self-directed dose change is recommended here.
Check: No target range or treatment is invented.

Common misconceptions

- Digestion and absorption are separate processes, and different GI segments have different functions.
- Stoma output and dehydration thresholds are not universal; output must be interpreted with anatomy, baseline, symptoms, and local clinical advice.
- A single endocrine value does not independently establish a diagnosis or medication adjustment.
- The pancreas has digestive exocrine and hormone-secreting endocrine functions; those roles should not be confused.
- General U.S. or UK educational references do not define Philippine clinical protocols.

Practice patterns

- For digestion questions, distinguish breakdown from absorption and identify the relevant tract segment.
- For diarrhea or stoma output, assess the pattern and fluid-status cues together; do not invent a threshold or replacement plan.
- For endocrine findings, connect symptom, timing, test context, and plan; avoid treating one value as the whole diagnosis.

Official scope references

Official scope: Enhanced Tables of Specifications for the Subjects in the Nurses Licensure Examination — Professional Regulation Commission; Professional Regulatory Board of Nursing — https://www.prc.gov.ph/sites/default/files/2025-10%20%20published%20Enhanced%20TOS%20nuse.pdf
Official scope: Nursing Licensure Examination Program, August 2026 — Professional Regulation Commission; Professional Regulatory Board of Nursing — https://www.prc.gov.ph/sites/default/files/exam%20progam%20aug%202026%20nle%20(nursing)%20signed.pdf

Instructional references

Instructional reference: Anatomy and Physiology 2e — OpenStax, Rice University — https://openstax.org/details/books/anatomy-and-physiology-2e
Instructional reference: 15.6 Applying the Nursing Process — Nursing Fundamentals 2e — Open RN; Wisconsin Technical Colleges System — https://wtcs.pressbooks.pub/nursingfundamentals/chapter/15-6-applying-the-nursing-process/
Instructional reference: Complications of an ileostomy — National Health Service, United Kingdom — https://www.nhs.uk/tests-and-treatments/ileostomy/complications/

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